



## Glossary terms from module 2

### **Terms and definitions from Course 7, Module 2**

Algorithm: A set of instructions for solving a problem or accomplishing a task

Boolean: A data type that has only two possible values, usually true or false

Branching: The ability of a program to alter its execution sequence

Comparator: An operator that compares two values and produces Boolean values (True/False)

def: A keyword that defines a function at the start of the function block

Docstring: A string at the beginning of a function's body that summarizes the function's behavior and explains its arguments and return values

elif: A reserved keyword that executes subsequent conditions when the previous conditions are not true

else: A reserved keyword that executes when preceding conditions evaluate as False

Function: A body of reusable code for performing specific processes or tasks

if: A reserved keyword that sets up a condition in Python

Logical operator: An operator that connects multiple statements together and performs complex comparisons

Modularity: The ability to write code in separate components that work together and that can be reused for other programs

Modulo: An operator that returns the remainder when one number is divided by another

Refactoring: The process of restructuring code while maintaining its original functionality

return: A reserved keyword in Python that makes a function produce new results which are saved for later use

Reusability: The capability to define code once and using it many times without having to rewrite it

Self-documenting code: Code written in a way that is readable and makes its purpose clear

## **Additional terms and definitions**

### **A**

Argument: Information given to a function in its parentheses

Assignment: The process of storing a value in a variable

Attribute: A value associated with an object or class which is referenced by name using dot notation

### **C**

Cells: The modular code input and output fields into which Jupyter Notebooks are partitioned

**Class:** An object's data type that bundles data and functionality together

**Computer programming:** The process of giving instructions to a computer to perform an action or set of actions

## **D**

**Data type:** An attribute that describes a piece of data based on its values, its programming language, or the operations it can perform

**Dot notation:** How to access the methods and attributes that belong to an instance of a class

**Dynamic typing:** Variables that can point to objects of any data type

## **E**

**Explicit conversion:** The process of converting a data type of an object to a required data type

**Expression:** A combination of numbers, symbols, or other variables that produce a result when evaluated

## **F**

**Float:** A data type that represents numbers that contain decimals

## **I**

**Immutable data type:** A data type in which the values can never be altered or updated

**Implicit conversion:** The process Python uses to automatically convert one data type to another without user involvement

Integer: A data type used to represent whole numbers without fractions

## **J**

Jupyter Notebook: An open-source web application for creating and sharing documents containing live code, mathematical formulas, visualizations, and text

## **K**

Keyword: A special word in a programming language that is reserved for a specific purpose and that can only be used for that purpose

## **M**

Markdown: A markup language that lets the user write formatted text in a coding environment or plain-text editor

Method: A function that belongs to a class and typically performs an action or operation

## **N**

Naming conventions: Consistent guidelines that describe the content, creation date, and version of a file in its name

Naming restrictions: Rules built into the syntax of the language itself that must be followed

## **O**

Object: An instance of a class; a fundamental building block of Python

Object-oriented programming: A programming system that is based around objects which can contain both data and code that manipulates that data

## **P**

Programming languages: The words and symbols used to write instructions for computers to follow

## **S**

String: A sequence of characters and punctuation that contains textual information

Syntax: The structure of code words, symbols, placement, and punctuation

## **V**

Variable: A named container which stores values in a reserved location in the computer's memory